COMPUTER SCIENCE + COMPUTER ENGINEERING

06

Module 23: Lighting, materials and real-time image quality

Expanded Volume 05 / four-lesson module

4 lessons · 20 lesson questions · 5 review questions · Integrating project

PACKAGE 06 / EXPANDED VOLUME 05 / SEPTEMBER 28, 2026
Self-study: derive, implement, test and explain.

Module 23 — Lighting, materials and real-time image quality

Prerequisites: Modules 07, 09-10, 21-22.

Planning allocation: 90-140 planning hours, including the project's work and review.

Lesson	Outcome
23.1 — Radiometry, reflectance and physically based materials	Explain how lighting models approximate light transport.
23.2 — Textures, mipmaps, filtering and tangent-space detail	Control aliasing while preserving the meaning of texture data.
23.3 — Shadows, light selection and rendering-path tradeoffs	Choose a lighting architecture from scene and bandwidth requirements.
23.4 — HDR, tone mapping, temporal reconstruction and validation	Separate display mapping from reconstruction and simulation correctness.

Original module project

Create a material-and-lighting demonstration with a documented color pipeline. Compare image quality and frame cost at identical resolution and scene content; include reference images, not only attractive screenshots.

Integrating milestones

1. P23-A: linear-light diffuse and documented microfacet material.
2. P23-B: semantic mip filtering and tangent-handedness tests.
3. P23-C: one shadow map and an equal-quality lighting-path comparison.
4. P23-D: exposure, tone mapping and temporal failure/rejection sequences.

Evidence to retain

- Units/space table; raw linear fixtures; material grid; normal seam test.
- Buffer budget; light-count data; shadow-bias comparison; temporal sequence and error domain.

Read the four lessons, keep the assigned readings, implement the workbook labs and use the module review before advancing. The reference code is a diagnostic aid; milestone completion is learner work.

LESSON 23.1 / MODULE 23

Radiometry, reflectance and physically based materials

> **Learning outcome:** Explain how lighting models approximate light transport.

Name the quantity before choosing the formula

Radiant flux is power, measured in watts. Irradiance is incident power per area, measured in W/m^2. Radiance adds a directional density, measured in W/(m^2 sr). These distinctions keep a light's intensity, a surface's reflectance and the displayed pixel value from becoming interchangeable sliders. In this volume, RGB lighting is a fixed-primary approximation to spectral transport, not a claim to represent every spectrum exactly. [S13]

A bidirectional reflectance distribution function, or BRDF, relates incident radiance from one direction to reflected radiance toward another direction at a surface. Its units are inverse steradians. The rendering equation integrates incoming radiance times the BRDF and the incident cosine over the hemisphere, with emission added separately. For an ideal Lambertian reflector of reflectance rho, f=rho/pi. The pi normalizes the cosine-weighted hemisphere integral; dropping it changes energy, not just artistic brightness. [R23]

Build one explicit material model

Start with diffuse reflectance, then add a single documented microfacet model. This package uses the common GGX normal distribution, Schlick Fresnel and separable Smith masking-shadowing as a teaching model. Let r be perceptual roughness and alpha=max(rr,*0.001). Let h be the normalized halfway vector of the unit light and view directions. With nh=max(dot(n,h),0), D=alpha^2/[pi*(nh^2(alpha^2-1)+1)^2]. The specular BRDF is DGF/(4*nl*nv) for positive nl and nv. At or below the surface, return zero instead of dividing by a tiny or negative denominator. [S14]

Schlick's approximation uses F=F0+(1-F0)*(1-vh)^5. For an illustrative dielectric with index of refraction 1.5 against air, normal-incidence reflectance is ((1.5-1)/(1.5+1))^2=0.04. Do not make 0.04 a universal material constant. Metals generally require colored specular response and do not use the same diffuse lobe as the simple dielectric route. A metallic workflow blends parameters under a stated convention; its sliders are not arbitrary color operations. [S03]

Normals and frames are part of the material

Transform normals using the inverse transpose of the linear model transform when needed, then normalize. Nonuniform scaling is the counterexample that exposes using the position matrix directly. Tangent frames, UV handedness and light directions must all agree about space. Decide whether a quantity is world-space, view-space or tangent-space before taking a dot product.

Plausibility is not validation

Test zero light, zero reflectance, normal incidence, grazing angles and roughness limits. Guard nonfinite inputs and an undefined halfway vector when view and light oppose one another. Clamp only at documented model boundaries; clamping the final HDR result to one hides missing exposure handling and can conceal energy mistakes.

A microfacet implementation can be physically motivated without being physically exact. The simple single-scattering model can lose energy for rough surfaces; it is not a complete multiple-scattering solution. A

white-furnace experiment must use a correctly integrated environment and isolate the model's assumptions. Do not claim energy conservation because one shiny sphere looks convincing. Preserve raw linear values alongside the display image.

Worked example

For rho=0.5 and irradiance E=10 W/m^2, an ideal diffuse reflector produces outgoing radiance L=rho*E/pi, approximately 1.59155 W/(m^2 sr), independently of view direction for this idealized model.

For the teaching specular model at normal incidence with roughness r=0.5, alpha=0.25. Therefore D=1/(pi*0.25^2), approximately 5.09296. With G=1 and F0=0.04, the specular BRDF is 5.09296*0.04/4, approximately 0.05093 sr^-1. D exceeding one is not by itself an energy violation; it is a density whose relevant integral matters.

As a geometry test, a normal (1,1,0) under nonuniform scale diag(2,1,1) should transform proportionally to (0.5,1,0), then be normalized. Transforming it as a position direction gives (2,1,0), a different orientation. Verify the transformed normal remains perpendicular to transformed tangent directions.

Original lab assignment

Implement diffuse lighting and one documented microfacet model. Pass: test a constant environment, grazing angles and roughness limits, and state approximations rather than labeling any shiny shader physically exact.

Assigned study

R23: Radiometry, Reflection Models and Light Transport; R19: shading.

Focused study: S13: radiometric quantities. S14: microfacet distributions. S03: metallic-roughness parameter semantics. R23: diffuse and light transport.

Check your understanding

1. What does a BRDF relate?
2. What is the Lambertian BRDF for reflectance rho?
3. Why document roughness encoding?
4. For rho=0.5 and E=10, compute ideal diffuse outgoing radiance.
5. Why does a BRDF or normal-distribution value above one not automatically imply energy creation?

LESSON 23.2 / MODULE 23

Textures, mipmaps, filtering and tangent-space detail

> **Learning outcome:** Control aliasing while preserving the meaning of texture data.

Textures are sampled signals

A texture is a discrete signal evaluated over a surface footprint. When many texels project onto one pixel, selecting a single nearest sample does not summarize that footprint; it aliases. Bilinear interpolation smooths between nearby samples but does not by itself provide the required large-footprint prefilter. A mip pyramid stores progressively filtered resolutions so minifying samples can use a coarser representation. [S15]

A full square power-of-two pyramid has N^2+(N/2)^2+...+1 texels. The geometric series approaches 4/3 of the base size. This factor is about texel count under those assumptions; row alignment, block compression, array layers and implementation allocation rules can change actual memory. Count the real level dimensions when sizes are not powers of two.

Filter according to meaning

Decode color into the chosen linear-light working representation before averaging it for mip generation. Then encode or store the result according to the runtime texture format. Averaging nonlinear color bytes changes the represented energy. Conversely, roughness, occlusion and normals are not color and should not receive an automatic sRGB decode. Normal-vector filtering may require renormalization and a policy for the lost subpixel distribution; renormalization alone is not a full antialiasing model for specular response. [S03, S15]

Alpha needs its own contract. Straight-alpha and premultiplied-alpha content require different filtering and compositing treatment. For cutout foliage, ordinary averaging can change apparent coverage at distant mip levels. Document whether the experiment preserves alpha coverage; do not silently call any averaged mask equivalent.

A tangent frame has handedness

A tangent-space normal is decoded, transformed by the tangent/bitangent/normal frame and normalized. If tangent t and geometric normal n are orthogonalized, a common bitangent construction is b=sign*cross(n,t), where sign records handedness. Mirrored UV islands can flip that sign. A tangent basis created by one convention and a normal map baked using another can produce a seam despite individually valid vectors. [S03]

Degenerate UV parameterizations cannot provide a meaningful inverse mapping for a tangent basis. Detect them and choose a documented fallback rather than dividing by a near-zero UV determinant. Include a mirrored-UV seam and nonuniformly scaled object in the validation scene.

Isotropic and anisotropic footprints

A receding floor produces a long, thin texture footprint. An isotropic mip choice must compromise between aliasing along one direction and unnecessary blur along the other. Anisotropic filtering samples a more elongated footprint within hardware and sampler limits. It is a quality/cost decision, not a guarantee that every high-frequency pattern can be perfectly reconstructed.

Begin with nearest and bilinear sampling, then mip selection, then compare supported anisotropy. Keep camera, resolution, texture and motion identical. Record the selected level or footprint estimate in a debug view. A nicer still screenshot can hide shimmer during motion, so include a fixed camera path as well as a single image.

Worked example

A 256-by-256 base texture has 65,536 texels. The full chain is 65,536+16,384+4,096+1,024+256+64+16+4+1=87,381 texels. At four bytes per texel, that is 349,524 payload bytes, before alignment or metadata. The exact factor is 87,381/65,536, slightly less than 4/3.

For color samples black and white, linear averaging gives 0.5, which becomes approximately 0.7354 under sRGB encoding. For a normal channel storing 0.5, the intended decoded signed component may be zero; applying sRGB decoding first would yield about 0.214 and then approximately -0.572 after remapping, an entirely different vector component.

For n=(0,0,1), t=(1,0,0), cross(n,t)=(0,1,0). A mirrored island with sign=-1 needs b=(0,-1,0). Omitting that sign flips how the normal map's second axis is interpreted.

Original lab assignment

Render a receding textured floor with and without mipmaps and inspect a normal-mapped seam. Pass: report texture storage including mip levels and explain why non-color data needs its own filtering policy.

Assigned study

R23: Textures and Sampling and Reconstruction; R21: texture images, samplers and mipmaps.

Focused study: S15: image-texture filtering and mipmaps. S03: color/data and tangent conventions. R23: sampled reconstruction.

Check your understanding

1. What problem do mipmaps primarily address?
2. What is the approximate total mip-chain storage factor?
3. Why can mirrored UVs break normal mapping?
4. How many texels and RGBA8 payload bytes are in the full 256-by-256 mip chain?
5. For n=(0,0,1), t=(1,0,0) and handedness -1, what bitangent is required?

LESSON 23.3 / MODULE 23

Shadows, light selection and rendering-path tradeoffs

> **Learning outcome:** Choose a lighting architecture from scene and bandwidth requirements.

A shadow map is another visibility computation

Render depth from the light's viewpoint. For a visible camera sample, transform its position into the same light clip convention and compare its depth with the stored light-visible depth. Samples outside the light's coverage need an explicit policy, not an accidental clamped texture lookup. A shadow map is finite-resolution evidence of visibility from one light, not a complete solution to global illumination.

Self-shadowing can arise from finite precision and different sampling of the receiver. A small depth bias can reduce this acne, but excessive bias detaches the shadow from the object, often called peter-panning. The sign and scale depend on the chosen depth convention, comparison and projection. Slope-sensitive bias and normal offsets address different geometric effects. First test a flat receiver and a small hovering object so the tradeoff is visible. [R19]

Percentage-closer filtering averages nearby depth-comparison results. It does not simply blur the depth texture and then compare once; those operations are not equivalent. Cascades, temporal stabilization and advanced soft-shadow methods are extensions after the basic comparison passes. A directional-light shadow should not swim merely because a camera moved a tiny amount without an explained projection change.

Choose where lighting work happens

In a simple forward renderer, geometry is drawn and shaded with selected lights directly into the color target. Its simplicity is useful for the small engine. In deferred shading, a geometry pass stores surface attributes in a G-buffer and a later pass performs lighting. That can reduce repeated shading for certain scenes, but introduces storage and bandwidth and complicates some combinations of transparency, material diversity and multisampling.

Tiled or clustered light selection assigns candidate lights to screen-space or screen/depth regions. It can benefit either forward or deferred rendering by reducing irrelevant light evaluations. Its cost includes building the lists, storing them, managing overflow and looking them up. Scene-dependent rejection is the mechanism; the word clustered is not itself a performance result. [R19]

Build a cost model you can test

Estimate pixels, light candidates, render-target bytes and passes. Then measure the declared scene. For each path, hold resolution, camera, visible content, light set and quality constant. A path that quietly drops lights is not a fair speed comparison. Overflow handling is part of correctness: a fixed-size list must either use a supported fallback, spill safely or report a documented quality reduction.

Track transparent geometry separately. A deferred opaque pass does not automatically solve ordered translucent composition. Keep the initial scope small: opaque objects, one directional shadow and a controlled point-light set. Add a transparent test only with a declared path and limitations.

Interpret the evidence carefully

Separate CPU list-building time from GPU lighting time and full-frame time. Report distributions, not only one number. A simplified arithmetic bandwidth estimate is a planning bound, not measured memory traffic; caches, compression, tiling and implementation details matter. The module's outcome is the ability to justify a renderer configuration and demonstrate its limits, not to prove that one architecture wins every scene.

Worked example

Suppose a 1920-by-1080 G-buffer stores 16 bytes per sample across its chosen attachments. One complete logical write is 1920*108*016=33,177,600 bytes, about 31.64 MiB. At 60 frames per second, that alone is roughly 1.99 GB/s of logical payload. Reading it once for lighting doubles this simplified traffic estimate; depth, final color, cache behavior and compression are not included.

With 256 lights and 2,073,600 visible pixels, naive all-lights shading evaluates about 530.8 million pixel/light candidates before rejection inside the shader. If clustering reduces the average candidate set to 12, the analogous count is 24.9 million, plus list-building and lookup overhead. That is a hypothesis for measurement, not a guaranteed 21-fold speedup.

For conventional increasing depth, compare receiver_depth-bias <= shadow_depth. Increasing a positive bias makes the test more permissive and can detach contact shadows. Reverse depth needs a correspondingly consistent rule rather than copying this sign blindly.

Original lab assignment

Implement shadow mapping and compare a simple forward path with tiled light selection or a small deferred prototype. Pass: disclose buffer sizes, light counts, transparency limitations and measured frame costs.

Assigned study

R19: real-time rendering and shadows; R23: Lights and Light Transport.

Focused study: R19: shadow mapping and real-time rendering paths. R23: lights and visibility. S07: depth-write to shader-read dependency for a native shadow map.

Check your understanding

1. Why can excessive shadow bias be visible?
2. Does deferred rendering remove bandwidth cost?
3. What determines whether clustered lighting helps?
4. What is the logical payload of a 1920-by-1080, 16-byte-per-pixel G-buffer write?
5. A tile light list has capacity 64 but receives 80 lights. What makes overflow handling an engineering requirement rather than an optional optimization?

LESSON 23.4 / MODULE 23

HDR, tone mapping, temporal reconstruction and validation

> **Learning outcome:** Separate display mapping from reconstruction and simulation correctness.

HDR storage and HDR display are different claims

An internal high-dynamic-range buffer can store scene values above one without implying that the application drives an HDR display. The first engine can render into floating-point linear RGB, apply exposure and a tone mapper, and present a standard-dynamic-range sRGB output. State this precisely. Clamping intermediate lighting to [0,1] destroys highlight information before the display mapping can use it.

Exposure scales the scene-linear value; an exposure change of e stops multiplies it by 2^e. A simple teaching tone mapper is T(x)=x/(1+x) for nonnegative x. It compresses highlights but is not a complete photographic or color-appearance model. Applying it independently per RGB channel can shift chromatic relationships. Document that limitation instead of naming the result physically correct. Output transfer encoding comes after tone mapping, exactly once. [R23]

Temporal reuse needs correspondence

Temporal accumulation combines a current sample with history, often Hnew=(1-w)*C*+*w*Hreprojected. The history must represent the same visible surface or signal. Reprojection uses current and previous transforms or motion vectors under an explicit pixel/UV convention. Camera motion and object motion are distinct sources; omitting either can put history on the wrong object.

A camera cut breaks the correspondence. Disocclusion reveals surfaces whose useful history may not exist. Resolution changes, projection changes, large exposure changes and changes to jitter policy can also invalidate or require transforming history. Compare depth, identity or other available signals conservatively, and reject or reduce history when correspondence is not credible. An old sample is not inherently better because it was expensive to compute. [R19]

Jitter and motion are not interchangeable

Subpixel camera jitter varies sample locations so multiple frames can gather additional information. Motion vectors describe movement of the represented surface between frames. Their conventions must account for whether jitter is included or excluded. Incorrectly mixing these transforms can cause a static scene to blur or crawl. Begin with an unjittered static accumulation experiment, then add a known subpixel pattern and a controlled camera path.

Neighborhood clipping or clamping can suppress ghosting but can also bias the result or remove detail. Treat it as a heuristic with visible tradeoffs. A basic temporal experiment is not equivalent to a production reconstruction system, frame generation or a vendor upscaler.

Validate space and time

A still image may look cleaner while motion produces long trails. Preserve both a fixed-frame image comparison and a repeatable sequence containing a camera cut, fast object motion, disocclusion and thin geometry. Compare to a more heavily sampled reference when possible. State whether error is measured in linear scene values, tone-mapped linear values or encoded display values; the choice changes the meaning of the number.

A useful test makes history obviously wrong: current value 0.2, old value 1.0 and a camera cut. With history weight 0.9, accidental reuse yields 0.92. Correct rejection yields 0.2. This numerical fixture cannot validate all visual quality, but it decisively detects a missing reset path.

Worked example

For scene-linear luminance x=4 and exposure -1 stop, the exposed value is 4*2^-1=2. The teaching tone mapper gives 2/(1+2)=2/3. sRGB encoding gives approximately 0.8360. Calling 2/3 the final encoded pixel would skip the output transfer step.

For C=0.2, H=1.0 and history weight w=0.9, the blend is 0.10.2+0.91=0.92. At a camera cut, reject history so the result is C=0.2. With a valid static history of H=0.21, the same weight gives 0.209, illustrating why correspondence determines whether averaging is useful.

For a 10-by-10 RGB comparison with absolute channel errors totaling 6.0, mean absolute error is 6/(10103)=0.02. Label the color domain and reference image; without them, the number is not reproducible or interpretable.

Original lab assignment

Add exposure, a documented tone mapper and a basic temporal accumulation experiment. Pass: capture a camera cut and disocclusion test, show failure modes, and compare both still-image error and motion behavior.

Assigned study

R23: Film and Imaging and Sampling; R19: anti-aliasing and image processing.

Focused study: R23: film, imaging and sampling. R19: antialiasing and reconstruction. S03: distinguish display color encoding from data handling.

Check your understanding

1. Are tone mapping and correct light transport the same task?
2. Why reject history at a camera cut?
3. Why are still screenshots insufficient for temporal methods?
4. Apply -1 exposure stop and T(x)=x/(1+x) to scene value 4. What value reaches the output transfer encoder?
5. Current sample is 0.2, history 1.0, weight 0.9. What are the blended value and correct camera-cut value?

Module 23 review

1. Why distinguish radiance from irradiance in a shader notebook?
2. Why can a perfect normal-map seam on one mesh be insufficient?
3. Why is a lower light-list count not a measured speedup?
4. Does an HDR intermediate prove HDR display output?
5. Why preserve camera-cut and disocclusion failures in the report?

Project handoff

Retain the material and lighting laboratory evidence. Explain one failure you deliberately induced and why the fix addresses its cause. Resolve review gaps before treating this module's outputs as reliable inputs to the next one.

Reading references: IDs retain their meanings in source/REFERENCES.md and the full handbook. Use those links alongside the lesson-specific reading assignments.